# GABBER'S THEOREM FOR THE ÉTALE TOPOLOGY, INTERNALLY

Abstract. In [1] Gabber's theorem is proved, in gerbe form and in $H^2$ form, in synthetic algebraic geometry with the *root topology*, generated by the roots of all monic polynomials of positive degree. Here we prove it for the étale topology, generated by Zariski covers and the roots of monic *unramifiable* polynomials. The étale theorem is not proved again: it is reduced to the root theorem. Two facts make the reduction work. First, an étale gerbe is a root gerbe: its paths, which are étale sheaves and étale torsors, are root sheaves by Hilbert 90. Second, an étale stack gerbe is a root stack gerbe: descent for objects along a root cover follows from Hilbert 90 applied to a product of path torsors over that cover. The sheafification for the étale topology is constructed, by trees, so the theorem has no hypothesis besides the axioms. The whole argument is checked in Unit.

## 1. Two topologies

We work, as in [1], in univalent type theory with propositional truncation $\|-\|$, over a local ring $R$ satisfying the axioms of synthetic algebraic geometry. A *topology* is given by a family of propositions $(G_j)_{j\in J}$; a type $X$ is a *$J$-sheaf* if for each $j$ the constant map $c : X \to (G_j \to X)$ is an equivalence. Sheaves are closed under $\Pi$, $\Sigma$, identity types and retracts, and being a sheaf is a proposition.

*The root topology.* For a monic $f$ of degree $d \geq 1$ let $G_f = \|\{x \mid f(x) = 0\}\|$. The root topology is generated by all such $G_f$; its sheafification is written $\Diamond$ ("locally"), and is constructed by towers of root adjunctions [1, Prop. 1.1]. Zariski covers need no generator: $R$ is local, so a family $g_1, \ldots, g_n$ generating 1 has merely an invertible member.

*The étale topology.* Let $B = R[X]/(f)$, with power basis $1, x, \ldots, x^{d-1}$, and $u = f'(x) \in B$. We say that $f$ is *unramifiable* if the $d$ coordinates of $u^d$ in the power basis generate the unit ideal of $R$. This is the condition of [2], §1, that the discriminant-like element $\Delta(f) = D(\delta_1, \ldots, \delta_d)$ be 1, where the $\delta_k$ are the coefficients of the characteristic polynomial of $u$: at a prime $\mathfrak{p}$, all $\delta_k$ lie in $\mathfrak{p}$ iff $u$ is nilpotent in $B \otimes k(\mathfrak{p})$, a $k(\mathfrak{p})$-algebra of dimension $d$, iff $u^d = 0$ there, iff all coordinates of $u^d$ lie in $\mathfrak{p}$. The *étale topology* is generated by the $G_f$ with $f$ unramifiable. By [2], Thm. 3.1, its covers are the classical étale covers, refined by trees of localisations and standard étale algebras. Every étale generator is a root generator, so every root sheaf is an étale sheaf; the converse fails.

**Definition 1.1.** An *étale tree* over a type $P$ is one of

- a leaf, given by an element of $P$;
- a root node: an unramifiable $f$ and, for every root $x$ of $f$, an étale tree over $P$;
- a Zariski node: $g_1, \ldots, g_n \in R$ with merely some $g_i$ invertible, and for each $i$ with $g_i$ invertible an étale tree over $P$.

Let $\Diamond_{\mathrm{et}} P = \|\text{an étale tree over } P\|$.

All data are small, so $\Diamond_{\mathrm{et}} P$ lies in the universe of $P$.

**Proposition 1.2.** *$\Diamond_{\mathrm{et}} P$ is a proposition and an étale sheaf, $P \to \Diamond_{\mathrm{et}} P$, and $\Diamond_{\mathrm{et}} P \to Q$ for every proposition $Q$ (in any universe) which is an étale sheaf and receives $P$. $\Diamond_{\mathrm{et}}$ is left exact.*

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*Proof.* A leaf gives $P \to \Diamond_{\mathrm{et}} P$. *Elimination*, by induction on the tree: at a root node, a map from the roots of $f$ to $Q$ factors through $G_f$, and $Q$ is an étale sheaf; at a Zariski node, $Q$ follows from the branch at an invertible $g_i$. *Sheaf*: $\Diamond_{\mathrm{et}} P$ is a proposition, so it suffices that $G_f \to \Diamond_{\mathrm{et}} P$ gives $\Diamond_{\mathrm{et}} P$ for $f$ unramifiable. The roots of $f$ are the points of $\operatorname{Spec} B$, $B = R[X]/(f)$ finitely presented, and at each point there is merely a tree. By $Z$-choice there merely are $h_1, \dots, h_n \in B$ generating 1 and, for each $i$ and each point $x$ with $h_i(x)$ invertible, a tree $t_{i,x}$. The tree with a root node $f$ followed, at the root $x$, by the Zariski node $h_1(x), \dots, h_n(x)$ with branches $t_{i,x}$, proves $\Diamond_{\mathrm{et}} P$; the $h_i(x)$ generate 1 at every point. *Left exactness* follows, as for any such closure, by gluing families of sheaves over $G_f$ as products [1, Prop. 1.1]. □

*Remark* 1.3. The Zariski node is inside the tree, after the root node, and in general it cannot be moved out: an étale tree need not be refined by a single tower of unramifiable root adjunctions followed by one Zariski cover (an example is $\sqrt{x}$ adjoined over $D(x)$, with 2 invertible, together with $D(1-x)$). For the root topology such a flattening exists, and the proof of the root theorem uses it. This is why the étale theorem is obtained by reduction to the root theorem.

Two facts are used repeatedly, for both topologies: a sheaf proposition that holds locally holds; and if $Q$ is a sheaf proposition with $G_j \to Q$, then $Q$.

## 2. Étale gerbes are root gerbes

A *band* at a point $x$ of a 1-type $X$ is a group isomorphism $\beta_x : (x = x) \to R^\times$. A *pre-torsor* is a set $T$ with an action of $R^\times$ which is free and transitive: for $t, t'$ in $T$ there is a unique $a$ with $at = t'$. In a 1-type with bands, each $a = b$ is a pre-torsor, $R^\times$ acting through the band [1].

**Definition 2.1.** An *étale gerbe* is a 1-type $X$ with a band at every point such that $\Diamond_{\mathrm{et}} X$, $\Diamond_{\mathrm{et}}(a = b)$ for all $a, b$, and each $a = b$ is an étale sheaf. It is an *étale stack gerbe* if moreover $X$ is an étale sheaf. *Root gerbes* and *root stack gerbes* are defined in the same way with $\Diamond$ and root sheaves [1].

**Lemma 2.2** (weakening)**.** $\Diamond_{\mathrm{et}} P \to \Diamond P$.

*Proof.* $\Diamond P$ is a root sheaf proposition, hence an étale sheaf, and receives $P$; apply Proposition 1.2. □

**Lemma 2.3** (Hilbert 90, étale)**.** *A pre-torsor $T$ which is an étale sheaf and satisfies $\Diamond_{\mathrm{et}} T$ is merely inhabited.*

*Proof.* As for the root topology [1]. Let $L$ be the module of maps $f : T \to R$ with $f(at) = a^{-1} f(t)$. It is a root sheaf, since $R$ is one, whatever $T$ is. At a point $t_0$, evaluation is an isomorphism $L \cong R$; so $\Diamond_{\mathrm{et}} T$, weakened to $\Diamond T$ (Lemma 2.2), gives that $L$ locally has a basis of rank one, and by descent of freeness for the root topology [1, Thm. 1.3] $L$ merely has a basis $f$. The type $\{t \mid f(t) = 1\}$ is a proposition: if $f(t) = f(t') = 1$ and $at = t'$ then $a^{-1} = 1$. It is an étale sheaf ($T$ is one, and $R$ is a root sheaf). At a point $t$ of $T$, $f(t)$ is a unit, since $f$ is a basis of $L \cong R$, and $f(f(t)\, t) = 1$. So $\Diamond_{\mathrm{et}} T$ gives a point with $f(t) = 1$. □

**Lemma 2.4.** *A merely inhabited pre-torsor is a root sheaf.*

*Proof.* Being a sheaf is a proposition, so we may take a point $t_0$. Then $T$ is a retract of $R^\times$ ($a \mapsto at_0$, and $t$ goes to the unique $a$ with $at_0 = t$), and $R^\times$ is a root sheaf. □

**Proposition 2.5.** *An étale gerbe is a root gerbe, with the same carrier and bands.*

*Proof.* $\Diamond X$ and $\Diamond(a = b)$ by Lemma 2.2. The pre-torsor $a = b$ is an étale sheaf and $\Diamond_{\mathrm{et}}(a = b)$, so it is merely inhabited (Lemma 2.3), hence a root sheaf (Lemma 2.4). □

**Theorem 2.6** (Gabber, gerbe form, étale)**.** *The gerbe form of Gabber's theorem* [1, Thm. 4.1] *holds for families of étale gerbes: given $A$ finitely presented, a family $G$ of étale gerbes over* $\operatorname{Spec} A$ *and an object of $k_*G_x$ for all $x$, merely there are $N$, a weight-one family $V_x$ on $G_x$ with fibres of rank $N$, and Azumaya algebras $E_x = \operatorname{End}(V_x)^{G_x}$ with $E_x \cong \operatorname{End}(V_{x,g})$ for $g : G_x$, such that $G_x \to \operatorname{Split}(E_x)$ is fully faithful and preserves the bands.*

*Proof.* Apply the root theorem to the root gerbes of Proposition 2.5. □

*Remark* 2.7. Each $E_x$ is étale-locally a matrix algebra: for $g : G_x$, $E_x \cong \operatorname{End}(V_{x,g}) \cong M_N(R)$ merely, and $\Diamond_{\mathrm{et}} G_x$.

## 3. Étale stack gerbes are root stack gerbes

The object-descent condition does not follow from Proposition 2.5: an étale sheaf need not be a root sheaf. For the carrier of an étale gerbe it does, again by Hilbert 90, now for the root topology. Fix a root generator $P = G_f$, a proposition. We need two lemmas about $P$.

**Lemma 3.1.** *Let $T : P \to$ pre-torsors. Then $\prod_{p:P} T_p$ is a pre-torsor, with the pointwise action.*

*Proof.* It is a set, and the action laws hold pointwise (function extensionality). Let $s, s'$ be in the product. For each $p$ there is a unique $\alpha_p$ with $\alpha_p s_p = s'_p$. Since $R^\times$ is a root sheaf, $\alpha : P \to R^\times$ is constant with a unique value $a_0$; then $a_0 s = s'$. If $bs = s'$, then $b = \alpha_p$ for each $p$ by uniqueness, so $\alpha$ is the constant map with value $b$, and $b = a_0$. The second components are paths in a set, hence equal. □

**Lemma 3.2.** *Let $X$ be a type whose identity types are root sheaves and $c : X \to (P \to X)$ the constant map. For $x_0 : X$ the fibre $\operatorname{Fib}(c\,x_0) = \sum_{x:X}(c\,x = c\,x_0)$ is contractible.*

*Proof.* For $q : c\,x = c\,y$, let $u(q) : x = y$ be the preimage of $\mathsf{happly}(q) : P \to (x = y)$ under the constant map of $x = y$, an equivalence. Then $\mathsf{ap}_c(u(q)) = q$: $\mathsf{happly}(\mathsf{ap}_c r)$ is the constant family $r$ (by induction on $r$), so $\mathsf{happly}(\mathsf{ap}_c(u(q))) = \mathsf{happly}(q)$, and $\mathsf{happly}$ is an equivalence. Hence $(x, q) \mapsto (x, u(q))$ and $(x, r) \mapsto (x, \mathsf{ap}_c r)$ exhibit $\operatorname{Fib}(c\,x_0)$ as a retract of $\sum_x (x = x_0)$, which is contractible. □

**Theorem 3.3** (descent for objects)**.** *The carrier of an étale stack gerbe is a root sheaf. Hence an étale stack gerbe is a root stack gerbe, with the same carrier and bands.*

*Proof.* Let $X$ be the carrier and fix $P = G_f$. We show that every fibre $\operatorname{Fib}(\varphi) = \sum_{x:X}(c\,x = \varphi)$ of $c : X \to (P \to X)$ is contractible; then $c$ is an equivalence.

*Reduction to a point.* $\operatorname{Fib}(\varphi)$ is an étale sheaf: $X$ is one, hence so are $P \to X$ and its identity types, and $\Sigma$ preserves sheaves. So is the proposition "$\operatorname{Fib}(\varphi)$ is contractible" ($\Sigma$, $\Pi$ and identity types again). As $\Diamond_{\mathrm{et}} X$, we may assume a point $x_0 : X$.

*Hilbert 90.* Let $T = \prod_{p:P}(x_0 = \varphi(p))$, a pre-torsor by Lemma 3.1. Its factors are root sheaves (Proposition 2.5), so $T$ is one. And $\Diamond T$: since $\Diamond T$ is a root sheaf, it suffices to prove it given $p : P$; then $P$ is contractible, $T \simeq (x_0 = \varphi(p))$, and $\Diamond_{\mathrm{et}}(x_0 = \varphi(p))$ gives $\Diamond(x_0 = \varphi(p))$ (Lemma 2.2). By Hilbert 90 for the root topology [1, §2], $T$ is merely inhabited.

*Conclusion.* The goal is a proposition, so we may take $s : T$. By function extensionality $s$ is a path $c\,x_0 = \varphi$, and along it $\operatorname{Fib}(\varphi)$ is $\operatorname{Fib}(c\,x_0)$, contractible by Lemma 3.2.

Finally the paths of $X$ are root sheaves and $\Diamond X$, $\Diamond(a = b)$ by Proposition 2.5. □

## 4. Gabber's theorem in $H^2$ form

Recall from [1] that $H^2(X)$ is the set truncation of $X \to \mathrm{StackGerbe}$, root stack gerbes in the first universe, with the contracted product as group law; $\delta(E)$ is the class of the family of splitting gerbes $\mathrm{Split}(E_x)$, and $\mathrm{cl}(G)$ the class of $G$.

**Theorem 4.1** (Gabber, $H^2$ form, étale)**.** *Let $A$ be finitely presented and $G$ a family of étale stack gerbes over* $\operatorname{Spec} A$*, and $k \geq 1$.*

(1) *If $k_*G_x$ has an object for every $x$, then merely there are $N$ and $E : \operatorname{Spec} A \to \mathrm{AZ}_N$ with $\delta(E) = \mathrm{cl}(G)$.*
(2) *If $k \cdot \mathrm{cl}(G) = 0$ in $H^2(\operatorname{Spec} A)$, the same conclusion holds.*

*Proof.* By Theorem 3.3, $G$ is a family of root stack gerbes; apply [1, Thm. 4.3] for (1) and [1, Thm. 5.6] for (2). □

*Remark* 4.2. In the proof of (1), $G_x = \mathrm{Split}(E_x)$ as stack gerbes, for every $x$; so $\mathrm{Split}(E_x)$ is itself an étale stack gerbe. The classes are compared in $H^2$ of root stack gerbes. Étale stack gerbes form a subtype of root stack gerbes, with the same identity types, since being an étale stack gerbe is a proposition about a root stack gerbe. We do not prove here that the induced map $H^2_{\mathrm{et}} \to H^2$ on set truncations is injective.

## 5. What is formalised in Unit

Everything above is checked in Unit, in `unit/sag/twisted`: `./segc sag/twisted all` gives 425 modules, 0 failed, about 11 s. No new axiom is added: the parameters are $R$ and the SAG axioms, with the existing Theory parameters univalence and replacement. Both sheafifications are constructed, not assumed. The local splitting of the remark after Theorem 2.6 and the comparison in the remark after Theorem 4.1 are not formalised, nor is the equivalence of our definition of unramifiable with $\Delta(f) = 1$, argued in Section 1.

| Statement | File |
|---|---|
| unramifiable polynomials | `Unramifiable` |
| étale trees, Proposition 1.2 | `EtaleTrees`, `EtaleTreeDescent`, `EtaleTreeClosure`, `ClosureLex` |
| Lemmas 2.2–2.4, Proposition 2.5 | `EtaleGabber` (`weaken`, `H90`, `Root`) |
| Theorem 2.6 | `EtaleGabber.Etale` |
| Lemma 3.1 | `EtaleStacks.PiTorsor` |
| Lemma 3.2, Theorem 3.3 | `EtaleStacks.Stack` |
| Theorem 4.1 (1), (2) | `EtaleStacks.Etale.H2`, `EtaleStacks.Etale.Torsion` |

The final declaration is

`EtaleStacks.Etale[R,setR,axioms].Torsion[A,fp,G,stk,n0,torsion].theorem`,

where `G` is a family of étale gerbes, `stk` says that each carrier is an étale sheaf, and `torsion` is an equality $(n_0 + 1) \cdot \mathrm{cl}(G) = 0$ in $H^2(\operatorname{Spec} A)$. A negative test is kept: an étale sheaf is not accepted as a root sheaf without proof.

## References

[1] *Gabber's theorem in $H^2$ form, internally*, `unit/doc/gabber-h2.pdf`, November 2026.
[2] T. Coquand, H. Lombardi, S. Neuwirth, Azumaya algebras and Barr's theorem, arXiv:2306.17679.
[3] O. Gabber, Some theorems on Azumaya algebras, in *The Brauer group*, LNM 844, Springer, 1981, 129–209.